Year 8

Mini Test 5

Algebra and Measurement

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Total Marks: 29 marks
Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Resources Allowed: 1 page A4 notes (1 side).

Question 1	**3 marks**

Complete the following sentences by using the following words. Note: there are more words than gaps. *coefficient, expanding, constant, factorising.*

a) Equations involving brackets can be solved by expanding ✓ the brackets and collecting like terms.

b) A number in front of a letter is known as a coefficient ✓

c) A number on its own in an expression is known as a constant ✓

Question 2	**2 marks**

Classify the following pairs as like terms (L) on not like terms (N) in order to add or subtract them.

a) 3a and 5a L ✓ $\frac{1}{2}$

b) 7x and -12x L ✓ $\frac{1}{2}$

c) 12ab and 4b N ✓ $\frac{1}{2}$

d) 2xy and -3yx L ✓ $\frac{1}{2}$

Question 3	**6 marks**

Simplify these algebraic expressions

a) 9m + 2m = 11m ✓ (1 mark)

b) 2a + 9 – 4a= -2a+9 ✓ (1 mark)

c) 4p x 3q = 12pq ✓ (1 mark)

d) 21a ÷ 3 = 7a ✓ (1 mark)

e) 7a + 4b – 4a + 2b= 3a+6b ✓ (1 mark)

f) $-18dg \div 4d=$ $\frac{-9g}{2}$ (1 mark)

Question 4	**6 marks**

Find the value of each of the following algebraic expressions given that a = 4 and b = -2.

a) $5a$ 20

b) $3b$ -6

c) $b - 5$ -7

d) $5a - 2b$ $20 - -4 = 24.$

e) $4a - 2a^2 + 10$ $16 - (2\times4) + 10. =$
$16 - 8 + 10 = 18.$

Question 5	**7 marks**

Expand and simplify where possible the following:

a) $2(x + 4) =$ $2x + 8$ (1 mark)

b) $-3(a - 5)=$ $-3a + 15$ (1 mark)

c) $3(m - 5) + 2m$ $3m - 15 + 2m = 5m - 15$ (2 marks)

d) $4(2d - 1) - 3(4e + 3d)$ $8d - 4 - 12e - 9d.$ (3 marks)
$= -d - 4 - 12e$

Question 6	**5 marks**

Factorise the following algebraic expressions:

a) $2b + 8$ $2(b + 4)$ (1 mark)

b) $4p + 6pq$ $2p(2 + 3q).$ (2 marks)

c) $-5xy^2 + 15yz$ $5y(-xy + 3z).$ (2 marks)